AN ISO 9001
CERTIFIED COMPANY

GENERAL OFFICES
2801 DAWSON ROAD
Tulsa< OK 74110
P: (918) 834-4611
F: (918) 832-0940
www.thecrosbygroup.com
crosbygroup@thecrosbygroup.com

August 2, 1993

TECHNICAL DATA SHEET

REUSE OF CROSBY SPELTER SOCKETS

The following are general guidelines for the reuse of Crosby 416/417 Spelter Sockets. The use and inspection of used spelter sockets is the responsibility of the user.

I. Procedure for Removing Spelter Core

A. Cut the rope close (1/2") to the nose of the socket and press the cone out of the basket.

B. We do not recommend the use of heat to remove the spelter cone for metallurgical, medical and environmental reasons.

1. However, if this is the only means available for removing the zinc cone, care should be taken not to exceed 850°F (450°C) surface temperature. The preferred method would be slow heat in a temperature controlled oven. If a torch (rosebud) is used, the heat spot shall be monitored with a tempil stick or a temperature indicator to prevent localized heating from exceeding the 850°F (450°C) limit.
2. To remove a **WIRELOCK**® cone, heat the surface of the socket to 350°F (do not exceed the 850°F limit for any localized hot spot). Leave for 5-10 minutes, then drive the cone out with a hammer and drift.

II. Selection of Sockets for Reuse

A. Use only sockets that:

- Do not show discoloration from excessive heating.
- Do not show any signs of welding.

B. Select only sockets that have been cleaned and have passed a Magnetic Particle Inspection by a qualified technician (Level II ASNT-SNT-TC-1A-88) per ASTM E709. Acceptance criteria shall be per ASTM E125, Types II-VIII, Degree 1. No cracks are acceptable.

C. Select only sockets that do not show any signs of overloading or wear on the socket or pin, (i.e. elongated pin holes, undersized pins, etc.).

D. Select sockets that are free from nicks, gouges and abrasions. Indications may be repaired by lightly grinding until surfaces are smooth, provided they do not reduce the dimensions by more than 10% of the nominal catalog dimension.

E. Select sockets that are not distorted, bent or deformed.

<u>Note: Sockets having any of the indications as outlined above (A-E) shall not be reused.</u>

III. Procedures for Speltering Sockets

A. The proper procedure for speltering sockets can be found on pages 112-116 of the *Wire Rope Users Manual*, 2nd Edition, and in *American Petroleum Institute (API) Recommended Practice 9B (RP9B)*, 9th Edition, May 30, 1986.

B. Some standards (API, ISO, BSI) recommend preheating of the zinc spelter socket before pouring. This temperature shall not exceed 850°F (450°C).

IV. Proof Testing

A. We recommend the socket assembly be proof tested at two (2) times the Working Load Limit (2 x WLL) assigned to the socketed assembly.

Copyright © 2014 The Crosby Group LLC
All Rights Reserved

WIRELOCK®

WARNINGS AND APPLICATION INSTRUCTIONS

⚠ WARNING

- **Incorrect use of WIRELOCK® can result in an unsafe termination which may lead to serious injury, death, or property damage.**
- **Do not use WIRELOCK® with stainless steel rope in salt water environment applications.**
- **Use only soft annealed iron wire for seizing.**
- **Do not use any other wire (copper, brass, stainless, etc.) for seizing.**
- **Never use an assembly until the WIRELOCK® has gelled and cured.**
- **Remove any non-metallic coating from the broomed area.**
- **Non Crosby sockets with large grooves need to have those grooves filled before use with WIRELOCK®.**
- **Read, understand, and follow these instructions and those on product containers before using WIRELOCK®.**

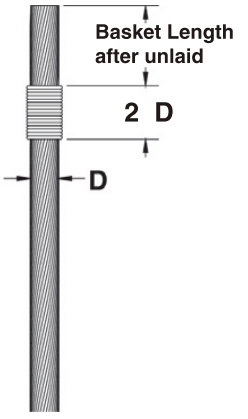
The following simplified, step-by-step instructions should be used only as a guide for experienced, trained users. For full information, consult our document ***WIRELOCK® TECHNICAL DATA MANUAL, API (AMERICAN PETROLEUM INSTITUTE) RECOMMENDED PRACTICE 9B, ISO standards, WIRE ROPE MANUFACTURERS CATALOGS,*** **and** ***WIRE ROPE SLING USERS MANUAL, THIRD EDITION.***

STEP 1 – SOCKET SELECTION

1. **WIRELOCK®** is recommended for use with Crosby 416-417 Spelter Sockets. Structural strand requires a socket with the basket length approximately five (5) times the strand diameter or fifty (50) times the wire diameter, what ever is greater to achieve 100% efficiency. Consult The Crosby catalog for proper selection of Wire Rope or Structural Strand sockets.
2. For use with sockets other than Crosby 416-417, consult the socket manufacturer or Crosby Engineering.
3. Sockets used with **WIRELOCK®** shall comply with Federal or International (CEN, ISO) Standards.
4. **WIRELOCK®**, as with all socketing media, depends upon the wedging action of the cone within the socket basket to develop full efficiency. A rough finish inside the socket may increase the load at which seating will occur. Seating is required to develop the wedging action.

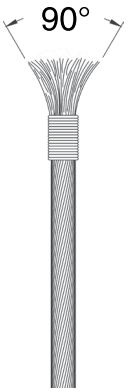
STEP 2 – MEASURE AND SEIZE

The rope ends to be socketed should be of sufficient length so that the end of the unlaid wires (from the strands) will be at the top of the socket basket. Seizing should be placed at a distance from the end equal to the length of the basket of the socket.



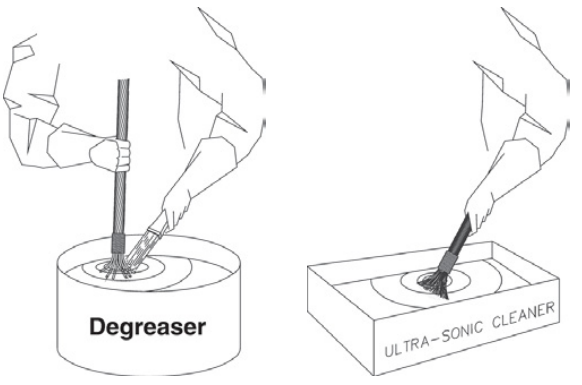
STEP 3 – BROOMING

1. Unlay the individual strands and fully broom out the wires of the wire rope and IWRC as far as the seizing. The wires should be separated but not straightened.
2. Cut out any fiber core.
3. Unlay the individual wires from each strand, including the IWRC, completely down to the seizing.
4. Remove any plastic material from broomed area.



STEP 4 – CLEANING

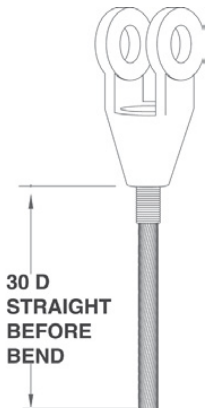
1. The method of cleaning will depend on the lubricant and/or coating on the wire.
2. The methods and materials used for cleaning should comply with the current EPA regulations.
3. Consult your Wire Rope supplier or Wire Rope manufacturer for recommended material and methods. Follow the solvent supplier's recommendations for cleaning the broomed end.



Copyright © 2014 The Crosby Group LLC
All Rights Reserved

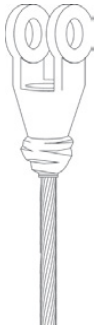
STEP 5 – POSITIONING OF SOCKET

1. Position socket over the broom until it reaches the seizing on the wire rope. The wires should be LEVEL with the top of the socket basket.
2. Clamp rope and socket vertically ensuring alignment of their axes.
3. **CAUTION: DO NOT USE OVERSIZED SOCKETS FOR WIRE ROPE.**



STEP 6 – SEAL SOCKET

Seal the base of the socket with putty or plasticine to prevent leakage of the **WIRELOCK®**.



STEP 7 – WIRELOCK® KITS

1. **WIRELOCK®** kits are pre-measured and consist of two (2) containers – one (1) with resin and one (1) with granular compound.
2. Use the complete kit – **NEVER MIX LESS THAN THE TOTAL CONTENTS OF BOTH CONTAINERS.**
3. Each kit has a shelf life clearly marked on each container and this must be observed. **NEVER USE OUT OF DATE KITS.**

⚠ CAUTION

- **WIRELOCK® resin, in liquid state, is flammable.**
- **Chemicals used in this product can give off toxic fumes and can burn eyes and skin.**
- **Never use out-of-date material.**
- **Use only in well-ventilated work areas.**
- **Never breathe fumes directly or for extended time.**
- **Always wear safety glasses to protect eyes.**
- **Always wear gloves to protect hands.**
- **Avoid direct contact with skin anywhere.**

STEP 8 – MIXING AND POURING

1. Mix and pour **WIRELOCK®** within the temperature range of 48 degrees to 95 degrees F. Booster kits are available for reduced temperatures.
2. Wirelock is set up to gel in 20 minutes @ 65°F. For every 18° rise in temperature the gel time will halve. At 83°F the gel time will be 10 minutes and at 101°F it will be 5 minutes. To give extra working time or pot life it is worth considering refrigerating the kits for two hours priot to mixing and pouring. The temperature of the socket will affect the cure time as well. The socket should also be as cool as possible - out of direct sunlight, as an example.
3. Pour all the resin into a container containing all the granular compound and mix thoroughly for two (2) minutes with a flat paddle.
4. The **WIRELOCK®** will turn a green blue color. If it does not turn a green blue color after mixing, DO NOT USE.
5. Immediately after mixing, slowly pour the mixture down one side of the socket until the socket basket is full.
6. Check for leakage at nose of socket, add putty if required.





STEP 9 – CURING



1. **WIRELOCK®** will gel in approximately fifteen (15) minutes, in a temperature range 65 degrees F to 75 degrees F.
2. The socket must remain undisturbed in the vertical position for an additional ten (10) minutes after gel is complete.
3. The socket will be ready for service sixty (60) minutes after gelling.
4. Never heat sockets to accelerate gel or curing.

STEP 10 – RE-LUBRICATION

Re-lubricate wire rope as required.

STEP 11 – PROOF LOADING

Whenever possible, the assembly should be proof loaded in accordance with ASME B30.9-2.6.

ALTERNATE SEIZING AND BROOMING METHOD

Reference the ***WIRELOCK® TECHNICAL DATA MANUAL*** from Crosby for an alternative socketing method.

Copyright © 2014 The Crosby Group LLC
All Rights Reserved

Crosby®

The Crosby S-423T Super TERMINATOR™ is the first wedge socket designed to take advantage of the performance properties associated with high performance, high strength, compacted strand, rotation resistant wire rope.

The Super TERMINATOR™ offers several advantages over traditional methods of wedge socket terminations:

- The innovative design will significantly increase the termination efficiency over existing wedge sockets available today.
- Terminations on most ropes have a minimum efficiency rating of 80% of the rope's catalog breaking strength.
- Patent pending design eliminates the difficulty of installing high performance, high strength, compacted strand, rotation resistant wire rope into a wedge socket termination.
- Proper application of the Super TERMINATOR™ eliminates the "first load" requirement of conventional wedge socket terminations.

Additional features:

- Wire rope sizes available: 5/8" -1 1/4", 14 mm- 32 mm
- Available as a complete assembly, or as a wedge kit that can be retrofitted onto existing Crosby S-421T TERMINATOR™ wedge sockets.
- Wedge accessories provided with a zinc finish.
- Meets or exceeds all ASME B30.26 requirements including: identification, ductility, design factor, proof load, and temperature requirements. Importantly, they meet other critical performance criteria not addressed by ASME B30.26 including: fatigue life, impact properties and material traceability.
- Available with bolt, nut and cotter (S-423TB)

The Super TERMINATOR™ by Crosby. The first wedge socket termination designed specifically for high performance wire rope.

Crosby®

www.thecrosbygroup.com



Wedge Socket Terminations

S-421T



US-422T



S-423T



Table of Contents

Wedge Socket Application and General Information 98-99

S-421T Wedge Sockets 100

US-422T Utility Wedge Sockets 101

Wedge Socket Warnings and Application Instructions 102-103

S-423T SUPER TERMINATOR™ 104

S-423T SUPER TERMINATOR™ Warnings and Application Instructions 105

Copyright © 2014 The Crosby Group LLC
All Rights Reserved

Wedge Socket Information

Application and General Information



S-421T
"Terminator"
Figure 1



US-422T
Utility
Figure 2



S-432T "Super
Terminator"
Figure 3

Wedge sockets are commonly used on the end of crane ropes as a means to attach the hoisting rope to the crane block or overhaul ball. Primary advantages of wedge style sockets include:

- Simplicity of installing and detaching product from the end of wire rope.
- When a mechanical spliced eye or poured socket is not practical.
- Applications where "on the job" attachment and quick or frequent rope replacement is necessary.

Crosby offers two standard styles of wedge sockets:

- The TERMINATOR™ Wedge Socket for use with standard wire ropes.
- The Super TERMINATOR™ Wedge Socket designed for high performance, high strength, compacted strand, rotation resistant wire rope.

Efficiency

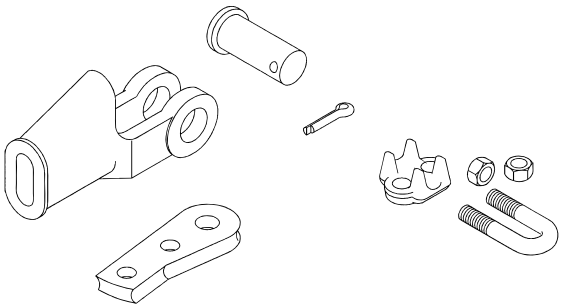
Efficiency ratings for wire rope terminations are based upon the minimum breaking force of wire rope.

S-421T and US-422T Sockets – The efficiency of Crosby Wedge Sockets is 80 percent when installed properly.

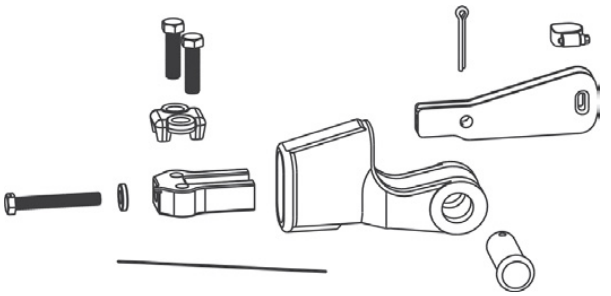
S-423T Super Terminator Sockets – Due to the unique construction of the various high performance, high strength, compacted strand, rotation resistant wire ropes, Crosby cannot make a broad general statement that all current and future designed ropes, when properly assembled with a Super TERMINATOR™ will achieve a minimum of 80 percent efficiency rating. To determine the efficiency rating for a specific rope, contact Crosby Engineering at 918-834-4611.
The rated load for wire rope assemblies such as wedge sockets is based on the following factors:

1. Wire rope minimum breaking force
2. 80 percent minimum connection efficiency
3. Design factor of the wire rope application

The schematic below depicts the parts of a typical Crosby S-421T and US-422T TERMINATOR™ Wedge Socket.



The schematic below depicts the parts of a typical Crosby S-423T "Super TERMINATOR™" Wedge Socket.



Key Facts About Wedge Sockets

1. S-421T and US-422T Wedge Socket terminations have an efficiency rating of 80% based on the minimum breaking force of XXIP wire rope.
2. Meets or exceeds all requirements of ASME B30.26 including identification, ductility, design factor, proof load and temperature requirements. Importantly, these sockets meet other critical performance requirements including fatigue life, impact properties and material traceability, not addressed by ASME B30.26.
3. The intended purpose of the SUPER TERMINATOR™ is to offer a wedge socket termination, which when assembled properly with high performance, high strength, compacted strand, rotation resistant wire rope will achieve an 80% termination efficiency. Due to the unique construction of these ropes, Crosby cannot make a broad general statement that all current and future designed ropes, when properly assembled with a SUPER TERMINATOR™, will achieve a minimum 80% termination efficiency. (To determine the efficiency rating for a specific rope, contact Crosby Engineering at 918-834-4611.)
4. Type Approval and certification in accordance with ABS 2006 Steel Vessel Rules 1-1-17.7, and ABS Guide for Certification of Cranes.
5. Basket is cast steel and Individually Magnetic Particle Inspected.

Copyright © 2014 The Crosby Group LLC
All Rights Reserved

Wedge Socket Information

6. Pin diameter and jaw opening allows wedge and socket to be used in conjunction with closed swage and spelter sockets.
7. Crosby TERMINATOR™ Wedge Sockets secure the tail or "dead end" of the wire rope to the wedge, thus eliminating loss or "punch out" of the wedge.
8. The TERMINATOR™ Wedge eliminates the need for an extra piece of rope, and is easily installed.
9. The TERMINATOR™ Wedge eliminates the potential breaking off of the tail due to fatigue.
10. The tail, which is secured by the base of the clip and the wedge, is left undeformed and available for reuse.
11. Utilizes standard Crosby Red-U-Bolt® wire rope clip.
12. The 3/8" thru 1-1/8" standard S-421 wedge socket can be retrofitted with the new style TERMINATOR™ Wedge.
13. Available with bolt, nut and cotter pin.
14. U.S. patent 5,553,360, Canada patent 2,217,004 and foreign equivalents.
15. Meets the performance requirements of EN 13411-6:200.
16. The SUPER TERMINATOR™ may be purchased as a complete wedge socket assembly or the wedge assembly may be purchased for retrofit onto your Crosby S-421T wedge socket basket.
17. The Crosby S-423TW SUPER TERMINATOR™ Wedge is designed to be assembled only into the Crosby S-421T socket basket. For the 1-1/4" S-423TW, assemble only on to S-421T basket marked TERMINATOR™.
18. Do not mix and match wedges or pins between models or sizes.
19. The SUPER TERMINATOR™ Wedge Socket may also be used with standard 6 to 8 strand and rotation resistant wire rope (special wire rope constructions with 8 or more strands).

Standard Considerations When Utilizing Wedge Sockets

ASME B30.5

Mobile and Locomotive Cranes, Section 5-1.7.1(d) states "The design factor specified in paras. 5-1.7.1(a) through (c) shall be the total minimum breaking strength of all ropes in the system divided by the load imposed on the rope system when supporting the static weights of structure and crane load. The absence of a derating of the wire rope breaking strength due to termination efficiency in paragraph 5-1.7.1(d) indicates that the termination efficiency was considered in the establishment of the design factors.

The methodology used in ASME B30.5 is not without precedent. API specification 2C for Offshore Pedestal Mounted Cranes 6th Edition, states "Wire rope design factors are intended to account for end connector efficiency and total reeving system efficiency of 80% or greater."

ASME B30.5 requires that "Wire rope clips used in conjunction with wedge sockets shall be attached to the unloaded dead end of the rope only."

ASME B30.26

The wedge socket shall be assembled as recommended by the manufacturer or a qualified person.

Wedge socket materials shall be of sufficient strength such that failure of the wire rope will occur before failure of the wedge socket.

Before installing a wedge socket on plastic coated or plastic impregnated wire rope, consult the manufacturer or a qualified person.

Each new wedge socket body and wedge shall have forged, cast, or die stamped marking by the manufacturer to show:

- Name or trademark of manufacturer
- Size
- Model, if required to match wedge to body

Crosby wedge sockets exceed ASME B30.26 identification requirements.

The live end of wire rope in the wedge socket cavity shall be in alignment with the socket's pin.

The assembler shall match the proper wedge with the socket for the wire rope to be installed.

Wedges shall not be interchanged between different manufacturers' sockets or models.

The length of dead end tail of the wire rope shall be as required by the manufacturer or a qualified person.

The dead end of the wire rope shall not be secured to the live end of the wire rope such that it restricts the movement of the live end.

Shock loading should be avoided.

See ASME B30.26 for full information.

Copyright © 2014 The Crosby Group LLC
All Rights Reserved

S-421T Wedge Sockets

QUIC-CHECK®

"TA" TYPE APPROVED

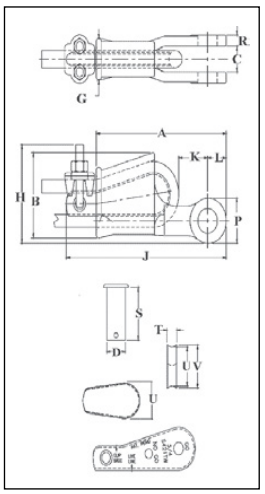
SEE APPLICATION AND WARNING INFORMATION

Para Español: www.thecrosbygroup.com *on Page 109*

S-421T



Wedge sockets meet the performance requirements of Federal Specification RR-S-550D, Type C, except those provisions required of the contractor. For additional information, see Crosby General Catalog.



- Wedge socket terminations have an efficiency rating of 80% based on the catalog strength of XXIP wire rope.
- Meets or exceeds all requirements of ASME B30.26 including identification, ductility, design factor, proof load and temperature requirements. Importantly, these sockets meet other critical performance requirements including fatigue life, impact properties and material traceability, not addressed by ASME B30.26.
- Type Approval and certification in accordance with ABS 2006 Steel Vessel Rules 1-1-17.7, and ABS Guide for Certification of Cranes.
- Basket is cast steel and individually Magnetic Particle Inspected.
- Pin diameter and jaw opening allows wedge and socket to be used in conjunction with closed swage and spelter sockets.
- Secures the tail or "dead end" of the wire rope to the wedge, thus eliminates loss or "Punch out" of the wedge.
- Eliminates the need for an extra piece of rope, and is easily installed.
- The TERMINATOR™ Wedge eliminates the potential breaking off of the tail due to fatigue.
- The tail, which is secured by the base of the clip and the wedge, is left undeformed and available for reuse.
- Incorporates Crosby's patented QUIC-CHECK® "Go" and "No-Go" feature cast into the wedge. The proper size rope is determined when the following criteria are met:

  1) The wire rope should pass thru the "Go" hole in the wedge.
  2) The wire rope should NOT pass thru the "No-Go" hole in the wedge.
- Utilizes standard Crosby Red-U-Bolt® wire rope clip.
- The 3/8" thru 1-1/8" standard S-421 wedge socket can be retrofitted with the new style TERMINATOR™ Wedge.
- Available with Bolt, Nut, and Cotter Pin.
- U.S. patent 5,553,360, Canada patent 2,217,004 and foreign equivalents.
- Meets the performance requirements of EN 13411-6: 2003.

S-421T Wedge Sockets
Assembly includes Socket, Wedge, Pin and Wire Rope Clip

Wire Rope Dia.		S-421T Stock No.	Weight Each (lbs.)	S-421TW Stock No. Wedge Only	Wedge Only Weight Each (lbs.)	Optional G-4082 Bolt, Nut & Cotter	
(in.)	(mm)					G-4082 Stock No.	Weight Each (lbs.)
3/8	9-10	1035000	3.18	1035555	.50	1092227	.38
1/2	11-13	1035009	6.15	1035564	1.05	1092236	.69
5/8	14-16	1035018	9.70	1035573	1.79	1092254	1.15
3/4	18-19	1035027	14.50	1035582	2.60	1092281	1.91
7/8	20-22	1035036	21.50	1035591	4.00	1092307	3.23
1	24-26	1035045	30.75	1035600	5.37	1092325	5.40
1-1/8	28	1035054	45.30	1035609	7.30	1092343	7.50
1-1/4	30-32	1035063	64.90	1035618	10.60	1092372	10.34

Wire Rope Dia.		S-421T Stock No.	Dimensions (in.)														
(in.)	(mm)		A	B	C	D	G	H	J*	K*	L	P	R	S	T	U	V
3/8	9-10	1035000	5.69	2.72	.81	.81	1.38	3.06	7.80	1.88	.88	1.56	.44	2.13	.44	1.25	1.38
1/2	11-13	1035009	6.88	3.47	1.00	1.00	1.62	3.76	8.91	1.26	1.06	1.94	.50	2.56	.53	1.75	1.88
5/8	14-16	1035018	8.25	4.30	1.25	1.19	2.12	4.47	10.75	1.99	1.22	2.25	.56	3.25	.69	2.00	2.19
3/4	18-19	1035027	9.88	5.12	1.50	1.38	2.44	5.28	12.36	2.41	1.40	2.63	.66	3.63	.78	2.34	2.56
7/8	20-22	1035036	11.25	5.85	1.75	1.63	2.69	6.16	14.37	2.48	1.67	3.13	.75	4.31	.88	2.69	2.94
1	24-26	1035045	12.81	6.32	2.00	2.00	2.94	6.96	16.29	3.04	2.00	3.75	.88	4.70	1.03	2.88	3.28
1-1/8	28	1035054	14.38	6.92	2.25	2.25	3.31	7.62	18.34	2.56	2.25	4.25	1.00	5.44	1.10	3.25	3.56
1-1/4	30-32	1035063	16.34	8.73	2.62	2.50	3.56	9.39	20.48	2.94	2.34	4.50	1.06	6.13	1.19	4.62	4.94

* Nominal **NOTE:** For intermediate wire rope sizes, use next larger size socket.

The S-423T Super TERMINATOR™ Wedge is designed to be assembled only into the Crosby S-421T TERMINATOR™ socket body.
IMPORTANT: The S-423TW for sizes 5/8" through 1-1/8" (14mm through 28mm) will fit respective size standard Crosby S-421T Basket. The 1-1/4" (30-32mm) S-423TW will only fit the Crosby S-421T 1-1/4" basket marked with " TERMINATOR™"

Copyright © 2014 The Crosby Group LLC
All Rights Reserved

US-422T Utility Wedge Sockets

QUIC-CHECK®

SEE APPLICATION AND WARNING INFORMATION

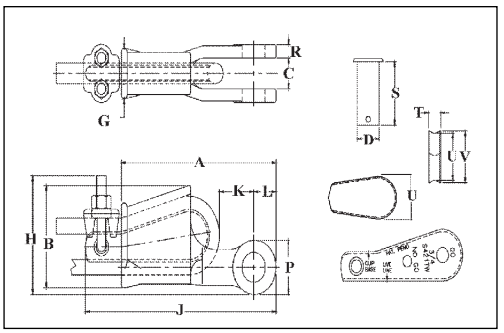
Para Español: www.thecrosbygroup.com *on Page 110*

US-422T



Most sizes now incorporate the "TERMINATOR™"design and may vary in shape from above product shown.

- Basket is cast steel and individually Magnetic Particle Inspected.
- Wedge socket terminations have an efficiency rating of 80% based on the catalog strength of XXIP wire rope.
- Wedges are color coded for easy identification.
  - Blue - largest wireline size for socket.
  - Black - mid size wireline for socket.
    - 7/16" on US4
    - 9/16" on US5
  - Orange - smallest wireline size for socket.
- Cast into each socket is the name "McKissick", "Crosby" or "CG", its model number and its wireline range.
- By simply changing out the wedge, each socket can be utilized for various wireline sizes (Ensure correct wedge is used for wire rope size).
- Cast into each wedge is the model number of the socket and the wireline size for which the wedge is to be used.
- Load pin is forged and headed on one end.
- US-422T Wedge Sockets contain a hammer pad (lip) to assist in proper securement of termination.
- Meets or exceeds all requirements of ASME B30.26 including identification, ductility, design factor, proof load and temperature requirements. Importantly, these sockets meet other critical performance requirements including fatigue life, impact properties and material traceability, not addressed by ASME B30.26.
- UWO-422T Wedges are to be used only with the US-422T Wedge Socket Assemblies.



US-422T Utility Wedge Sockets

Model No.	Wire Rope Size (in.)	Wire Rope Size (mm)	US-422T Stock No.	Weight Each (lbs.)	Wedge Only Stock No.	Weight Each (lbs.)	Dimensions (in.) A	B	C	D	G	H	J	K	L	P	R	S	T	U	V
US4T	3/8	10	1044300	4.6	1047310	.6	6.81	3.55	1.00	1.00	1.63	2.81	8.43	1.38	1.06	1.94	.50	2.53	.44	1.91	2.14
US4T	7/16	11	1044309	4.6	1047301	.6	6.81	3.55	1.00	1.00	1.63	2.81	8.73	1.08	1.06	1.94	.50	2.53	.53	1.76	1.88
US4T	1/2	13	1044318	4.6	1047329	.6	6.81	3.55	1.00	1.00	1.63	2.81	8.73	1.02	1.06	1.94	.50	2.53	.53	1.76	1.88
US5T	1/2	13	1044327	8.5	1047338	1.0	9.19	4.23	1.41	1.25	2.13	3.31	11.19	1.84	1.50	3.00	.63	3.25	.75	1.92	2.16
US5T	9/16	14	1044336	8.5	1047347	1.0	9.19	4.23	1.41	1.25	2.13	3.31	11.47	2.40	1.50	3.00	.63	3.25	.69	2.00	2.18
US5T	5/8	16	1044345	8.5	1047356	1.0	9.19	4.23	1.41	1.25	2.13	3.31	11.47	2.34	1.50	3.00	.63	3.25	.69	2.00	2.18
US6T	5/8	16	1044354	9.4	1047365	1.4	9.45	4.70	1.50	1.25	2.24	3.63	11.91	2.48	1.50	3.00	.56	3.25	.88	2.38	2.75
US6T	3/4	19	1044363	9.4	1047374	1.4	9.45	4.70	1.50	1.25	2.24	3.63	11.81	2.03	1.50	3.00	.56	3.25	.88	2.13	2.63
US8AT	5/8	16	1044372	19.8	1047383	4.3	10.59	5.68	1.81	1.63	2.38	5.53	13.19	1.91	1.53	2.88	.75	4.13	.69	3.26	3.50
US8AT	3/4	19	1044381	20.4	1047392	4.8	10.59	5.68	1.81	1.63	2.38	5.84	13.54	2.38	1.53	2.88	.75	4.13	.78	3.12	3.38
US7*	7/8	22	1038580	16.5	1046674	2.6	11.26	5.11	1.31	1.25	2.69	—	—	2.56	1.63	3.26	.66	3.25	1.06	2.12	2.56
US7*	1	25	1038589	16.5	1046683	2.6	11.26	5.11	1.31	1.25	2.69	—	—	2.56	1.63	3.26	.66	3.25	1.06	1.88	2.38
US8T	7/8	22	1044404	31.5	1047425	7.6	12.77	6.96	1.81	1.63	3.06	7.20	16.02	2.87	1.65	3.12	.75	4.13	.88	3.88	4.18
US8T	1	25	1044417	32.5	1047431	8.6	12.77	6.96	1.81	1.63	3.06	7.31	16.41	2.32	1.65	3.12	.75	4.13	1.03	3.76	4.06
US10T	1-1/8	28	1044426	55.4	1047440	12.5	15.94	8.62	1.81	1.63	3.57	9.15	19.72	3.26	2.19	4.38	.75	4.13	1.09	4.76	5.06
US10T	1-1/4	32	1044435	58.0	1047459	15.0	15.94	8.62	1.81	1.63	3.57	9.39	20.22	2.83	2.19	4.38	.75	4.13	1.19	4.62	4.94
US11T	1-1/8	28	1044444	60.6	1047468	12.5	16.34	8.73	2.62	2.50	3.56	9.15	19.97	3.37	2.34	4.50	1.06	6.13	1.09	4.76	5.06
US11T	1-1/4	32	1044453	64.9	1047477	15.0	16.34	8.73	2.62	2.50	3.56	9.39	20.48	2.94	2.34	4.50	1.06	6.13	1.19	4.62	4.94

* Non-"TERMINATOR"™ Style

Copyright © 2014 The Crosby Group LLC
All Rights Reserved

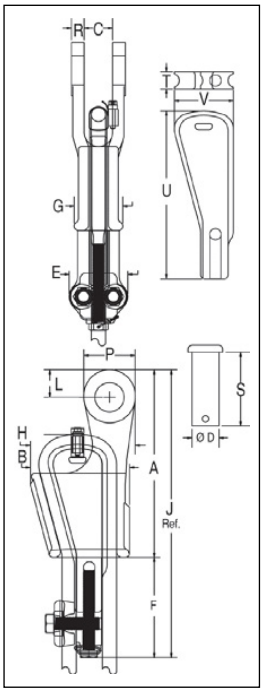
S-423T SUPER TERMINATOR™

SEE APPLICATION AND WARNING INFORMATION
Para Español: www.thecrosbygroup.com *on Page 111*

S-423T



Wedge sockets meet the performance requirements of Federal Specification RR-S-550D, Type C, except those provisions required of the contractor. Meets the performance requirements of EN13411-6:2003. For additional information, see Crosby General Catalog.



- Wedge socket terminations have a minimum efficiency rating on most high performance, high strength, compacted strand, rotation resistant wire ropes of 80% based on the catalog breaking strength of the various ropes.**
- Patent Pending design eliminates the difficulty of installing high performance wire rope into a wedge socket termination.
- Proper application of the Super TERMINATOR™ eliminates the "first load" requirement of conventional wedge socket terminations.
- S-423TW Wedge Kit can be retrofitted onto existing Crosby S-421T TERMINATOR™ Wedge Sockets.
- Wedge and accessories provided with a zinc finish.
- Meets the performance requirements of EN13411-6:2003.
- Meets or exceeds all requirements of ASME B30.26 including identification, ductility, design factor, proof load and temperature requirements. Importantly, these sockets meet other critical performance requirements including fatigue life, impact properties and material traceability, not addressed by ASME B30.26.
- Basket is cast steel and Individually Magnetic Particle Inspected.
- Pin diameter and jaw opening allows wedge and socket to be used in conjunction with closed swage and spelter sockets.
- Secures the tail or "dead end" of the wire rope to the wedge, thus eliminates loss or "punch out" of the wedge.
- Eliminates the need for an extra piece of rope, and is easily installed.
- The TERMINATOR™ Wedge eliminates the potential breaking off of the tail due to fatigue.
- The tail, which is secured by the base of the clip and the tension device, is left undeformed and available for reuse.
- Available with Bolt, Nut, and Cotter Pin.

**** NOTICE: Due to the unique construction of various ropes, Crosby cannot make a broad general statement that all current and future design of ropes, when properly assembled with the Super TERMINATOR™, will achieve a minimum 80% termination efficiency. Contact wire rope manufacturer or Crosby Engineering (918-834-4611) to determine efficiency rating for a specific rope.**

S-423T Wedge Sockets
Assembly includes Socket, Wedge, Pin, Wire Rope Clip, Tensioner, Bolts and Secondary Retention Wire.

Wire Rope Dia.		S-423T Assembly with Round Pin and Cotter Pin			S-423TB Assembly with Bolt, Nut and Cotter Pin			S-423TW** Wedge Kit		
		S-423T	S-423T Weight Each		S-423TB	S-423TB Weight Each		S-423TW	S423TW Weight Each	
(in.)	(mm)	Stock No.	(lbs.)	(kg)	Stock No.	(lbs.)	(kg)	Stock No.	(lbs.)	(kg)
5/8	14- 16	1035123	12.7	5.8	1035218	13.1	5.9	1034018	5.2	2.4
3/4	18-19	1035132	19.4	8.8	1035227	19.1	8.7	1034027	7.2	3.3
7/8	20-22	1035141	28.8	13.1	1035236	27.8	12.6	1034036	10.3	4.7
1	24-26	1035150	39.2	17.8	1035245	37.3	16.9	1034045	11.9	5.4
1-1/8	28	1035169	57.1	25.9	1035254	57.9	25.9	1034054	19.9	9.0
1-1/4	30-32	1035178	88.6	40.2	1035272	88.1	39.9	1034063	33.8	15.3

** Kit contains Wedge, Wire Rope Clip and Bolts, Tensioner, Tensioner Bolt and Secondary Retention Wire.

Wire Rope Dia.		S-423T Stock No.	Dimensions (in.)															
(in.)	(mm)		A	B	C	D	G	F	G	H	J*	L	P	R	S	T	U	V
5/8	14-16	1035123	8.25	4.50	1.25	1.19	3.00	4.06	2.13	4.61	12.31	1.22	2.25	.56	3.25	.75	6.88	2.60
3/4	18-19	1035132	9.88	5.20	1.50	1.38	3.25	4.81	2.44	5.37	14.69	1.40	2.62	.66	3.63	.88	7.65	3.02
7/8	20-22	1035141	11.25	5.88	1.75	1.63	3.81	5.73	2.69	6.16	16.98	1.67	3.13	.75	4.31	1.00	9.47	3.47
1	24-26	1035150	12.81	6.56	2.00	2.00	3.81	5.73	2.94	7.05	18.54	2.01	3.75	.88	4.70	1.13	10.41	3.82
1-1/8	28	1035169	14.38	6.94	2.25	2.25	4.00	6.85	3.38	7.81	21.23	2.26	4.25	1.00	5.44	1.25	11.83	4.22
1-1/4	30-32	1035178	16.34	8.63	2.62	2.50	4.50	7.76	3.57	9.38	24.10	2.34	4.50	1.06	6.62	1.38	13.87	5.82

* Nominal

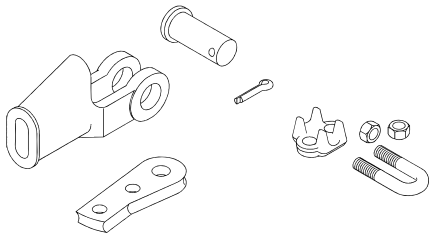
NOTE: For intermediate wire rope sizes, use next larger size socket.
The S-423T Super TERMINATOR™ wedge is designed to be assembled only into the Crosby S-421T TERMINATOR™ Socket Body.
IMPORTANT: The S-423TW for sizes 5/8" through 1-1/8" (14mm through 28mm) will fit respective size standard Crosby S-421T Basket. The 1-1/4" (30-32mm) S-423TW will only fit the Crosby S-421T 1-1/4" basket marked with " TERMINATOR™".

Copyright © 2014 The Crosby Group LLC
All Rights Reserved

WEDGE SOCKET

WARNINGS AND APPLICATION INSTRUCTIONS

Extended Wedge Socket Assembly U.S. Patent No. 5,553,360 and Canada Patent No. 2,217,004



S-421T / US-422T
"TERMINATOR"™

NOTE: The design of the basket for the S-421T 1-1/4" TERMINATOR™ Wedge Socket does not allow proper fit to the old style Crosby S-421W wedge (see Fig. 1). **Do not assemble or use.**

The design of the basket for each US-422T TERMINATOR™ Wedge Socket does not allow proper fit to the old style UWO-422 wedge (See Fig. 1). **Do not assemble or use.**

All S-421T and US-422T TERMINATOR™ baskets are marked with a capital "T" or TERMINATOR™.

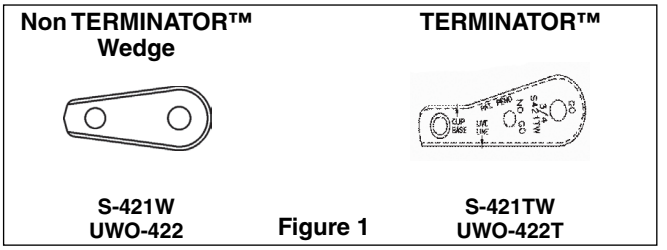


Figure 1

QUIC-CHECK® "Go" and "No-Go" features cast into wedge. The proper size wire rope is determined when the following criteria are met:

1. The wire rope shall pass thru the "Go" hole in the wedge.
2. The wire rope shall NOT pass thru the "No-Go" hole in the wedge.

QUIC-CHECK®

Important Safety Information – Read and Follow Inspection/Maintenance Safety

- Always inspect socket, wedge and pin before using.
- Do not use part showing cracks.
- Do not use modified or substitute parts.
- Repair minor nicks or gouges to socket or pin by lightly grinding until surfaces are smooth. Do not reduce original dimension more than 10%. Do not repair by welding.
- Inspect permanent assemblies annually, or more often in severe operating conditions.
- Do not mix and match wedges or pins between models or sizes.
- Always select the proper wedge and socket for the wire rope size.

Assembly Safety

- Use only with standard 6 to 8 strand wire rope of designated size. For intermediate size rope, use next larger size socket. For example: When using 9/16" diameter wire rope use a 5/8" size Wedge Socket Assembly. Welding of the tail on standard wire rope is not recommended. The tail length of the dead end should be a minimum of 6 rope diameters but not less than 6" (See Figure 2).
- **To use with Rotation Resistant** wire rope (special wire rope constructions with 8 or more outer strands) ensure that the dead end is welded, brazed or seized before inserting the wire rope into the wedge socket to prevent core slippage or loss of rope lay. The tail length of the dead end should be a minimum of 20 rope diameters but not less than 6" (See Figure 2).
- Properly match socket, wedge and clip (See Table 1) to wire rope size.
- Align live end of rope, with center line of pin.(See Figure 2)
- Secure dead end section of rope. (See Figure 2)
- Do not attach dead end to live end or install wedge backwards (See Fig. 3).
- **Use a hammer to seat Wedge and Rope as deep into socket as possible before applying first load.**

⚠ WARNING

- **Loads may slip or fall if the Wedge Socket is not properly installed.**
- **A falling load can seriously injure or kill.**
- **Read and understand these instructions before installing the Wedge Socket.**
- **Do not side load the Wedge Socket.**
- **Apply first load to fully seat the Wedge and Wire Rope in the socket. This load should be of equal or greater weight than loads expected in use.**
- **Do not interchange wedges between S-421T and US422T or between sizes.**
- **Do not assemble an old style 1-1/4" S-421W wedge into an S-421T 1-1/4" TERMINATOR™ basket.**
- **Do not assemble an old style UWO-422 Wedge into a US-422T TERMINATOR™ basket.**

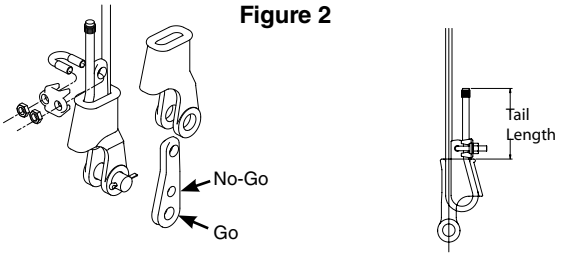


Figure 2

*Tail Length	
Standard 6 to 8 Strand Wire Rope	**Rotation Resistant Wire Rope**
A minimum of 6 rope diameters, but not less than 6"	A minimum of 20 rope diameters, but not less than 6"

TABLE 1								
Rope Size	3/8	1/2	5/8	3/4	7/8	1	1-1/8	1-1/4
Clip Size	3/8	1/2	5/8	3/4	7/8	1	1-1/8	1-1/4
* Torque Ft./lbs.	45	65	95	130	225	225	225	360

* The tightening torque values shown are based upon the threads being clean, dry, and free of lubrication.

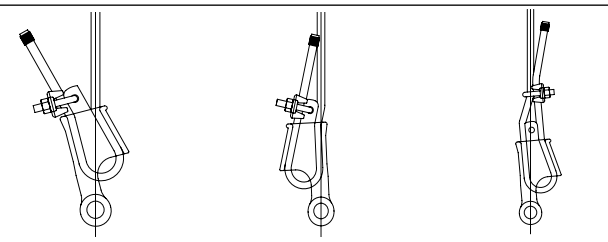


Figure 3

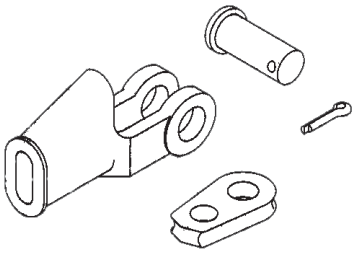
Operating Safety

- Apply first load to fully seat the Wedge and Wire Rope in the socket. This load should be of equal or greater weight than loads expected in use.
- Efficiency rating of the Wedge Socket termination is based upon the catalog breaking strength of Wire Rope. The efficiency of a properly assembled Wedge Socket is 80%.
- During use, do not strike the dead end section with any other elements of the rigging (called two blocking).

Copyright © 2014 The Crosby Group LLC
All Rights Reserved

WEDGE SOCKET

WARNINGS AND APPLICATION INSTRUCTIONS



S-421 / US-422

Important Safety Information - Read and Follow Inspection/Maintenance Safety

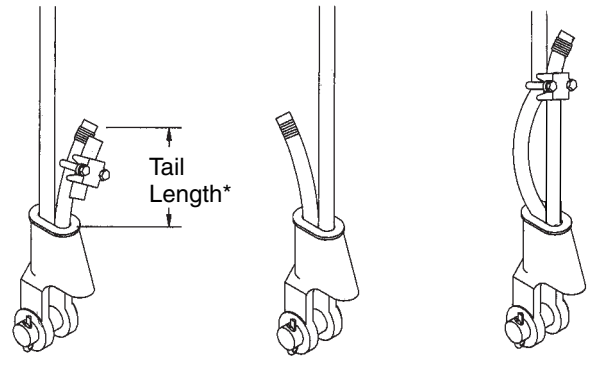
- Always inspect socket, wedge and pin before using.
- Do not use part showing cracks.
- Do not use modified or substitute parts.
- Repair minor nicks or gouges to socket or pin by lightly grinding until surface are smooth. Do not reduce original dimension more than 10%. Do not repair by welding.
- Inspect permanent assemblies annually, or more often in severe operating conditions.
- Do not mix and match wedges or pins between models or sizes.
- Always select the wedge and socket for the wire rope size.

Assembly Safety

- Use only with standard 6 to 8 strand wire rope of designated size. For intermediate size rope, use next larger size socket. For example: When using 9/16" diameter wire rope use a 5/8" size Wedge Socket Assembly. Welding of the tail on standard wire rope is not recommended. The tail length of the dead end should be a minimum of 6 rope diameters but not less than 6".
- Align live end of rope, with center line of pin. (See Figure 1)
- Secure dead end section of rope. (See Figure 1)
- DO NOT ATTACH DEAD END TO LIVE END. (See Figure 1)
- Use a hammer to seat Wedge and Rope as deep into socket as possible before applying first load.
- To use with Rotation Resistant wire rope (special wire rope constructions with 8 or more outer strands) ensure that the dead end is welded, brazed or sized before inserting the wire rope into wedge socket to prevent core slippage or loss of rope lay.The tail length of the dead end should be a minimum of 20 rope diameters but not less than 6". (Figure 1)

⚠WARNING

- **Loads may slip or fall if the Wedge Socket is not properly installed.**
- **Load misapplied in direct contact with the wedge can dislodge the wedge & cause loss of load.**
- **A falling load can seriously injure or kill.**
- **Read and understand these instructions before installing the Wedge Socket.**
- **Do not side load the wedge socket.**
- **Do not interchange Crosby wedge socket, wedge or pin with non Crosby Wedge socket, wedge or pin.**
- **Apply first load to fully seat the Wedge and Wire Rope in the socket. This load should be of equal or greater weight than loads expected in use.**
- **Do not interchange wedge between S-421 and US-422 or between sizes.**



*Tail Length
Standard 6 to 8 strand wire rope
A minimum of 6 rope diameters, but not less than 6"
(i.e. - For 1" rope: Tail Length = 1" x 6 = 6")

Rotation Resistant Wire Rope
A minimum of 20 rope diameters, but not less than 6"
(i.e. - For 1" rope: Tail Length = 1" x 20 = 20")

Figure 1

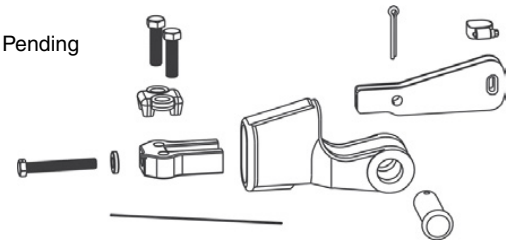
Operating Safety

- Apply first load to fully seat the Wedge and Wire Rope in the socket. This load should be of equal or greater weight than loads expected in use.
- Efficiency rating of the Wedge Socket termination is based upon the catalog breaking strength of Wire Rope. The efficiency of properly assembled Wedge Socket is 80%.
- During use, do not strike the dead end section with any other elements of the rigging (called two-blocking).
- Do not allow a direct load to contact the wedge.

Copyright © 2014 The Crosby Group LLC
All Rights Reserved

SUPER TERMINATOR™ WEDGE SOCKET WARNING & APPLICATION INSTRUCTIONS

Patent Pending



S-423T "SUPER TERMINATOR™"

The intended purpose of the SUPER TERMINATOR™ is to offer a Wedge Socket termination, which when assembled properly with high performance, high strength, compacted strand, rotation resistant wire rope will achieve an 80% termination efficiency. Due to the unique construction of these ropes, Crosby cannot make a broad general statement that all current and future designed ropes, when properly assembled with a SUPER TERMINATOR™, will achieve a minimum 80% termination efficiency. (To determine the efficiency rating for a specific rope, contact Crosby Engineering at 918-834-4611.)

The SUPER TERMINATOR™ may be purchased as a complete Wedge Socket assembly or the wedge assembly may be purchased for retrofit onto your Crosby S-421T wedge socket basket.

The Crosby S-423TW SUPER TERMINATOR™ Wedge is designed to be assembled only into the Crosby S-421T socket basket. For the 1-1/4" S-423TW, assemble only on to S-421T basket marked TERMINATOR™.

Important Safety Information - Read and Understand

Inspection/Maintenance Safety

- Always inspect socket, wedge and pin before using.
- Do not use part showing cracks.
- Do not use modified or substitute parts.
- Repair minor nicks or gouges to socket or pin by lightly grinding until surfaces are smooth. Do not reduce original dimension more than 10%. Do not repair by welding.
- Inspect permanent assemblies annually, or more often in severe operating conditions.
- Do not mix and match wedges or pins between models or sizes.
- Always select the proper wedge and socket for the wire rope size.

Assembly Safety

- Properly match socket and wedge assembly to wire rope size.
- Ensure the dead end is properly seized before inserting the wire rope into the wedge socket basket. High performance, high strength, compacted strand, rotation resistant wire ropes are sensitive to seizing methods. For specific seizing procedures, contact the wire rope manufacturer.
- The tail length of the dead end should be a minimum of 20 rope diameters but not less than 10" (See Fig. 1).
- Mount wedge socket basket in vice.
- Insert live end of wire rope into wedge basket, aligning live end of rope with center line of pin. Make a loop and return. (See Figure 2).
- Pull on live line to remove excess out of loop, leaving enough room to properly insert wedge into basket. (See Figure 3).
- Secure rope to SUPER TERMINATOR™ Wedge with clamp (See Figure 4).
- Pull Wedge and rope into basket until tensioner bolt, with washers properly applied, can engage threads in nose of wedge. (See Figure 5).
- Use torque wrench to tighten tensioner bolt to recommended torque value, properly seating wedge and rope into basket. Reference Table 1 for recommended Torque in Ft. Lbs.
- Secure dead end section of rope with clip base. Tighten bolts to recommended torque values (See Table 1).
- Properly install wire to securely lock tensioner bolt to tensioner. (See Figure 6).
- Do not attach dead end to live end or install wedge backwards. (See Figure 7).

Operating Safety

- Proper application of the Super TERMINATOR™ eliminates the "first load" requirement of conventional wedge socket terminations.
- Efficiency rating of the Wedge Socket termination is based upon the catalog breaking strength of Wire Rope. The efficiency of a properly assembled Super Terminator on most high performance, high strength, compacted strand, rotation resistant ropes will achieve 80% of catalog breaking strength of rope, depending on the unique construction of these ropes. (To determine the efficiency rating for a specific rope, contact Crosby Engineering at 918-834-4611.)
- During use, do not strike the dead end section with any other elements of the rigging (Called two blocking).
- The SUPER TERMINATOR™ wedge socket may also be used with standard 6 to 8 strand and rotation resistant wire rope (special wire rope constructions with 8 or more strands).

⚠ WARNING

- **Loads may slip or fall if the Wedge Socket is not properly installed.**
- **A falling load can seriously injure or kill.**
- **Read and understand these instructions before installing the Wedge Socket.**
- **Do not side load the Wedge Socket.**
- **Apply recommended torque to tensioner and clip bolts, and properly install wire to securely lock tensioner bolt to tensioner.**
- **Do not assemble the S-423 Wedge in any brand or model socket basket other than the Crosby S-421T TERMINATOR™.**
- **The size is marked on the socket basket and wedge, do not interchange wedge between sizes.**

Figure 1



Tail Length
A minimum of 20 rope diameters, but not less than 10"

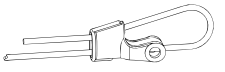


Figure 2

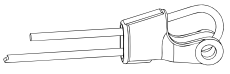


Figure 3

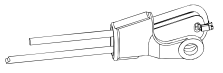


Figure 4

TABLE 1
S-423T Torque Value Table

Wedge Size (in.)	Tensioner Bolt Torque Ft./lbs*	Clip Bolts Torque Ft./lbs*
5/8	110	95
3/4	150	130
7/8	380	225
1	380	225
1-1/8	600	225
1-1/4	900	360

* The tightening torque values shown are based upon the threads being clean, dry, and free of lubrication.

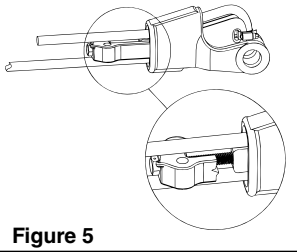


Figure 5

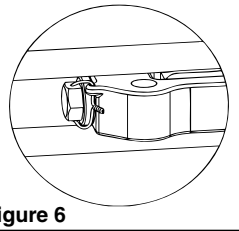
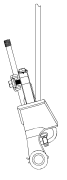


Figure 6

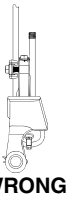
Figure 7



WRONG
Rope Backward



WRONG
Wedge Backward



WRONG
Tail Clipped to Live Line

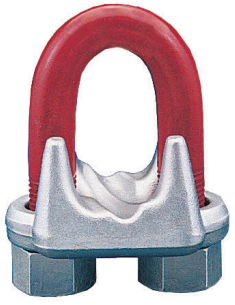
Copyright © 2014 The Crosby Group LLC
All Rights Reserved

Notes

Copyright © 2014 The Crosby Group LLC
All Rights Reserved

Wire Rope Clips

G-450
Wire Rope Clips



SS-450
Stainless Steel
Wire Rope Clips



New Style
Fist Grip® Clips
3/16" - 5/8"



Fist Grip® Clips
3/4" - 1-1/2"



Table of Contents

Wire Rope Clip Information..... 108-109

Forged Wire Rope Clips.....110

U-Bolt Clip Installation Procedures 111

Crosby Clips Warnings and Application Instructions 112

Fist Grip® Wire Rope Clips.....113

Fist Grip® Clip Installation Procedures 114

Crosby Fist Grip® Clips Warnings and Application Instructions 115

Copyright © 2014 The Crosby Group LLC
All Rights Reserved

Wire Rope Clip Information

Application and General Information



G-450 Crosby Red-U-Bolt® Clip

Figure 1



SS-450 Crosby Stainless Steel Clip

Figure 2



G-429 Crosby Fist Grip® Clip 3/16"-5/8"

Figure 3



G429 Crosby Fist Grip® Clip 3/4"-1-1/2"

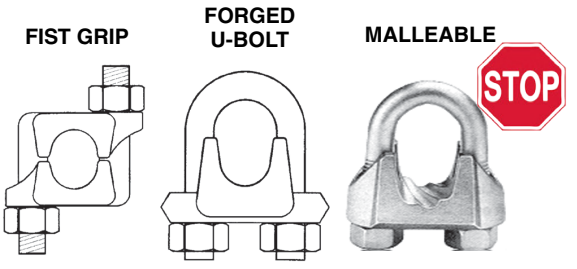
Figure 4

There are three basic types of wire rope clips – forged U-Bolt, forged fist grip, and cast malleable. Wire rope clips are sometimes referred to as Crosby's or clamps. Wire rope clips are used to make a loop at the end of wire rope.

- Crosby "U-Bolt" style clips consists of a U-shaped bolt, a forged saddle and two nuts (see Figure 1 & 2).
- Crosby Fist Grip® clips consists of two saddles and two nuts (see Figures 3 & 4).
- Malleable clips consist of a U-shaped bolt, an inferior cast iron saddle and two nuts.

It is worth noting that Crosby does not manufacturer cast malleable clips. **(IMPORTANT – Only the forged clips can be used for the critical applications such as guying, tie down and suspension.)**

WIRE ROPE CLIP TYPES
CRITICAL APPLICATIONS



"WIRE ROPE CLIPS SHALL BE DROP-FORGED STEEL OF SINGLE SADDLE (U-BOLT) OR DOUBLE SADDLE CLIP. MALLEABLE CAST IRON CLIPS SHALL NOT BE USED."

(POH651AA)

ASME B30.5

Key Facts About Wire Rope Clips

1. The proper performance of forged clips depends on proper manufacturing practices that include good forging techniques and accurate machining.
2. Forged clips ("U-bolt" and Fist Grip® styles) provide a greater rope bearing surface and more consistent strength than malleable cast iron clips.
3. Fist Grip® clips provide a saddle for both the "live" and the "dead end."
4. Fewer forged clips ("U-bolt" and Fist Grip® styles) are required for each termination than with malleable cast iron clips.
5. Forged clips ("U-bolt" and Fist Grip® styles) reduce the possibility of hidden defects that are sometimes present in malleable cast iron clips.
6. Malleable cast iron clips should only be used in non-critical applications. ASME, OSHA, and ASTM recommend only forged clips for critical applications.
7. All Clips are individually bagged or tagged with proper application instructions and warning information. User must become familiar with instructions attached to the product.
8. *Crosby wire rope clips can be reused after careful inspection of threads and saddle area. Proper torque values must be attained whenever Crosby clips are reused. The U-bolt must fit into the base without requiring a forceful change in U-bolt spread. The clip assembly must be properly installed and capable of being torqued to its full recommended value. The roddles in the clip base must be undamaged and the clip assembly must be re-torqued after the initial load is applied.
9. Whenever possible, use thimbles when forming eyes with wire rope clips. Wire rope thimbles were designed to provide the wire rope with protection against wear. They are not considered a primary load carrying member and are not rated by a load carrying capacity.

Efficiency

Efficiency ratings for wire rope end terminations are based upon the minimum breaking force of wire rope.

Eyes formed in wire rope using properly installed clips have the following efficiency rating.

Wire Rope Size	Efficiency
1/8" through 7/8"	80%
1" through 3-1/2"	90%

Copyright © 2014 The Crosby Group LLC
All Rights Reserved

Wire Rope Clip Information

The efficiency of the termination depends on:

1. Selection of the proper size clips
2. Utilizing the proper number of clips
3. Utilizing the proper amount of rope turned back
4. Correct placement of the clips
5. Applying the proper torque to the nuts

It is important to recognize that efficiency cannot be increased by adding extra clips, but it can be reduced by over tightening.

Standard Considerations When Utilizing Clips

OSHA 1025.251(c) 4 (iii)
Eyes in wire rope bridles, slings, or bull wires shall not be formed by wire rope clips or knots.

ASME B30.9
The use of wire rope clips to fabricate wire rope slings is generally prohibited. See ASME B30.9 for full details.

ANSI A17.1 AND ANSI A10.4.
For elevator, personnel hoist, and scaffold applications, refer to ANSI A17.1 and ANSI A10.4. These standards do not recommend U-Bolt style clips.

Maritime
The maritime standards are divided into the following categories: Ship Repairing, Shipbuilding, Ship Breaking, Longshoring, and Gear Certification (Vessel Gear and Shore-Based Gear). The use of wire rope clips to form eyes in wire rope and wire rope slings is allowed in Ship Repairing, Shipbuilding and Ship Breaking. In Longshoring Subpart F 1918.52 “Specific requirements” (a)(2) reads, “Wire rope clips or knots shall not be used to form eyes in or join sections of, preventer guys.” Also, 1918.52 “Specific Requirements” (c)(3) reads, “Eyes in the ends of wire rope cargo falls shall not be formed by knots and in single part falls, shall not be formed by wire rope clips.” In gear certification Subpart D 1919.24 “Limitations on use of wire rope” (c) reads, “Eyes in the ends of wire rope cargo falls shall not be formed by knots and in single part falls, shall not be formed by wire rope clips.”

ASME B30.26
Saddles shall be forged steel.

Each wire rope clip saddle shall have forged or die stamped markings by the manufacturer to show
1. Name or trademark of manufacturer
2. Size

Crosby clips exceed ASME B30.26 identification requirements.

Shock loading should be avoided per ASME B30.26.

See ASME B30.26 for full information.

Copyright © 2014 The Crosby Group LLC
All Rights Reserved

Forged Wire Rope Clips

SEE APPLICATION AND WARNING INFORMATION

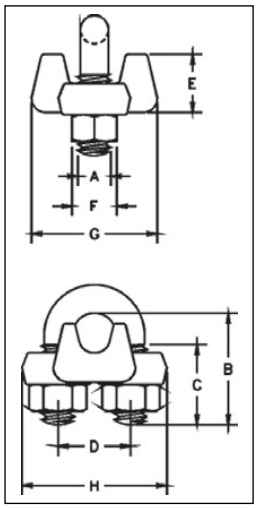
Para Español: www.thecrosbygroup.com *on Page 118*

G-450



- Each base has a Product Identification Code (PIC) for material traceability, the name CROSBY or CG, and a size forged into it.
- Based on the catalog breaking strength of wire rope, Crosby wire rope clips have an efficiency rating of 80% for 1/8″ - 7/8″ sizes, and 90% for sizes 1″ through 3-1/2″.
- Entire clip - Galvanized to resist corrosive and rusting action.
- Sizes 1/8″ through 2-1/2″ and 3″ have forged bases.
- All clips are individually bagged or tagged with proper application instructions and warning information.
- Clip sizes up through 1-1/2″ have rolled threads.
- Meets or exceeds all requirements of ASME B30.26 including identification, ductility, design factor, proof load and temperature requirements. Importantly, these wire rope clips meet other critical performance requirements including fatigue life, impact properties and material traceability, not addressed by ASME B30.26.
- Look for the Red-U-Bolt®, your assurance of Genuine Crosby Clips.

Crosby Clips, all sizes 1/4″ and larger, meet the performance requirements of Federal Specification FF-C-450 TYPE 1 CLASS 1, except for those provisions required of the contractor. For additional information, see the Crosby General Catalog.



G-450 Crosby® Clips

Rope Size (in.)	Rope Size (mm)	G-450 Stock No.	Std. Package Qty.	Weight Per 100 (lbs.)	Dimensions (in.) A	B	C	D	E	F	G	H
1/8	3-4*	1010015	100	6	.22	.72	.44	.47	.37	.38	.81	.99
3/16*	5*	1010033	100	10	.25	.97	.56	.59	.50	.44	.94	1.18
1/4	6-7	1010051	100	19	.31	1.03	.50	.75	.66	.56	1.19	1.43
5/16	8	1010079	100	28	.38	1.38	.75	.88	.73	.69	1.31	1.66
3/8	9-10	1010097	100	48	.44	1.50	.75	1.00	.91	.75	1.63	1.94
7/16	11	1010113	50	78	.50	1.88	1.00	1.19	1.13	.88	1.91	2.28
1/2	12-13	1010131	50	80	.50	1.88	1.00	1.19	1.13	.88	1.91	2.28
9/16	14-15	1010159	50	109	.56	2.25	1.25	1.31	1.34	.94	2.06	2.50
5/8	16	1010177	50	110	.56	2.25	1.25	1.31	1.34	.94	2.06	2.50
3/4	18-20	1010195	25	142	.62	2.75	1.44	1.50	1.39	1.06	2.25	2.84
7/8	22	1010211	25	212	.75	3.12	1.62	1.75	1.58	1.25	2.44	3.16
1	24-26	1010239	10	252	.75	3.50	1.81	1.88	1.77	1.25	2.63	3.47
1-1/8	28-30	1010257	10	283	.75	3.88	2.00	2.00	1.91	1.25	2.81	3.59
1-1/4	32-34	1010275	10	438	.88	4.44	2.22	2.34	2.17	1.44	3.13	4.13
1-3/8	36	1010293	10	442	.88	4.44	2.22	2.34	2.31	1.44	3.13	4.19
1-1/2	38	1010319	10	544	.88	4.94	2.38	2.59	2.44	1.44	3.41	4.44
1-5/8	41-42	1010337	Bulk	704	1.00	5.31	2.62	2.75	2.66	1.63	3.63	4.75
1-3/4	44-46	1010355	Bulk	934	1.13	5.75	2.75	3.06	2.92	1.81	3.81	5.24
2	48-52	1010373	Bulk	1300	1.25	6.44	3.00	3.38	3.03	2.00	4.44	5.88
2-1/4	56-58	1010391	Bulk	1600	1.25	7.13	3.19	3.88	3.19	2.00	4.56	6.38
2-1/2	62-65	1010417	Bulk	1900	1.25	7.69	3.44	4.13	3.69	2.00	4.69	6.63
** 2-3/4	** 68-72	1010435	Bulk	2300	1.25	8.31	3.56	4.38	4.88	2.00	5.00	6.88
3	75-78	1010453	Bulk	3100	1.50	9.19	3.88	4.75	4.44	2.38	5.31	7.61
** 3-1/2	** 85-90	1010426	Bulk	4000	1.50	10.75	4.50	5.50	6.00	2.38	6.19	8.38

* Electro-plated U-Bolt and Nuts. ** 2-3/4″ and 3-1/2″ base is made of cast steel.

SS-450



- Each base has a Product Identification Code (PIC) for material traceability, the name CROSBY or "CG", and a size forged into it.
- Entire clip is made from 316 Stainless Steel to resist corrosive and rusting action.
- All components are Electro-Polished.
- All clips are individually bagged or tagged with proper application instructions and warning information.

SS-450 Stainless Steel Wire Rope Clips

Rope Size (in.)	Rope Size (mm)	SS-450 Stock No.	Std. Package Qty.	Weight Per 100 (lbs.)	Dimensions (in.) A	B	C	D	E	F	G	H
1/8	3-4	1011250	Bulk	6	.22	.72	.44	.47	.41	.38	.81	.94
3/16	5	1011261	Bulk	10	.25	.97	.56	.59	.50	.44	.94	1.16
1/4	6-7	1011272	Bulk	20	.31	1.03	.50	.75	.66	.56	1.19	1.44
3/8	9-10	1011283	Bulk	47	.44	1.50	.75	1.00	.91	.75	1.63	1.94
1/2	12-13	1011305	Bulk	77	.50	1.88	1.00	1.19	1.13	.88	1.91	2.28
5/8	16	1011327	Bulk	106	.56	2.38	1.25	1.31	1.34	.94	2.06	2.50

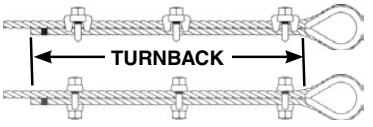
Copyright © 2014 The Crosby Group LLC
All Rights Reserved

Wire Rope Clip Information

U-Bolt Clip Installation Procedures

WIRE ROPE CLIPS

BASIC REQUIREMENTS FOR PROPER TERMINATION



SELECT PROPER SIZE CLIPS AND TURNBACK.

PLACE CLIPS ON IN PROPER SEQUENCE.

TORQUE ALL CLIPS EVENLY WITH A TORQUE WRENCH.

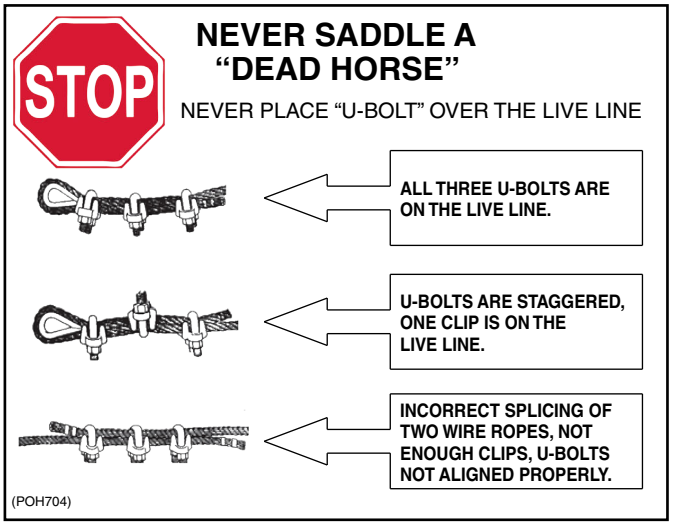
APPLY FIRST LOAD AND RE-TORQUE WITH A TORQUE WRENCH.

(POH651)

Turnback is measured from the end of the rope to the base of the eye or to the thimble. This part of the rope is often referred to as the dead end. Since the U-Bolt Clip has a single saddle, you must be concerned about workers saddling the dead horse.

The table below defines the information required for each wire rope clip to achieve maximum efficiency.

Table 1				
Clip Size (in.)	**Rope Size (in.)**	**Minimum No. of Clips**	**Amount of Rope to Turn Back in Inches**	***Torque in Ft. Lbs.**
1/8	1/8	2	3-1/4	4.5
3/16	3/16	2	3-3/4	7.5
1/4	1/4	2	4-3/4	15
5/16	5/16	2	5-1/4	30
3/8	3/8	2	6-1/2	45
7/16	7/16	2	7	65
1/2	1/2	3	11-1/2	65
9/16	9/16	3	12	95
5/8	5/8	3	12	95
3/4	3/4	4	18	130
7/8	7/8	4	19	225
1	1	5	26	225
1-1/8	1-1/8	6	34	225
1-1/4	1-1/4	7	44	360
1-3/8	1-3/8	7	44	360
1-1/2	1-1/2	8	54	360
1-5/8	1-5/8	8	58	430
1-3/4	1-3/4	8	61	590
2	2	8	71	750
2-1/4	2-1/4	8	73	750
2-1/2	2-1/2	9	84	750
2-3/4	2-3/4	10	100	750
3	3	10	106	1200
3-1/2	3-1/2	12	149	1200
If a pulley (sheave) is used for turning back the wire rope, add one additional clip. See Figure 4.				
If a greater number of clips are used than shown in the table, the amount of turnback should be increased proportionately.				
* The tightening torque values shown are based upon the threads being clean, dry and free of lubrication.				



The following example illustrates the correct use of Table 1 in determining the proper application of Crosby G-450 Red U-BOLT Clips.

Example: An installation requires the use of 3/8" 6 x 19 RRL, IPS wire rope.

From Table 1:

- Clip Size = 3/8"
- Number of Clips = 2
- Amount of Turnback = 6-1/2"
- Torque in Ft. Lbs. = 45

Table 1				
Clip Size (in.)	**Rope Size (in.)**	**Minimum No. of Clips**	**Amount of Rope to Turn Back in Inches**	***Torque in Ft. Lbs.**
1/8	1/8	2	3-1/4	4.5
3/16	3/16	2	3-3/4	7.5
1/4	1/4	2	4-3/4	15
5/16	5/16	2	5-1/4	30
3/8	3/8	2	6-1/2	45
7/16	7/16	2	7	65

Copyright © 2014 The Crosby Group LLC
All Rights Reserved

CROSBY® CLIPS

WARNINGS AND APPLICATION INSTRUCTIONS



G-450
(Red-U-Bolt®)



SS-450
(316 Stainless Steel)

⚠WARNING

- **Failure to read, understand, and follow these instructions may cause death or serious injury.**
- **Read and understand these instructions before using clips.**
- **Match the same size clip to the same size wire rope.**
- **Prepare wire rope end termination only as instructed.**
- **Do not use with plastic coated wire rope.**
- **Apply first load to test the assembly. This load should be of equal or greater weight than loads expected in use. Next, check and retighten nuts to recommended torque (See Table 1, this page).**

Efficiency ratings for wire rope end terminations are based upon the catalog breaking strength of wire rope. The efficiency rating of a properly prepared loop or thimble-eye termination for clip sizes 1/8" through 7/8" is 80%, and for sizes 1" through 3-1/2" is 90%.

The number of clips shown (see Table 1) is based upon using RRL or RLL wire rope, 6 x 19 or 6 x 36 Class, FC or IWRC; IPS or XIP, XXIP. If Seale construction or similar large outer wire type construction in the 6 x 19 Class is to be used for sizes 1 inch and larger, add one additional clip. If a pulley (sheave) is used for turning back the wire rope, add one additional clip.

The number of clips shown also applies to rotation-resistant RRL wire rope, 8 x 19 Class, IPS, XIP, XXIP sizes 1-1/2 inch and smaller; and to rotation-resistant RRL wire rope, 19 x 7 Class, IPS, XIP, XXIP sizes 1-3/4 inch and smaller.

For other classes of wire rope not mentioned above, we recommend contacting Crosby Engineering to ensure the desired efficiency rating.

For elevator, personnel hoist, and scaffold applications, refer to ANSI A17.1 and ANSI A10.4. These standards do not recommend U-Bolt style wire rope clip terminations. The style wire rope termination used for any application is the obligation of the user.

For OSHA (Construction) applications, see OSHA 1926.251.

1. Refer to Table 1 in following these instructions. Turn back specified amount of rope from thimble or loop.

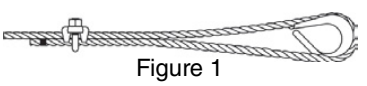


Figure 1

Apply first clip one base width from dead end of rope. Apply U-Bolt over dead end of wire rope – live end rests in saddle (Never saddle a dead horse!). Use torque wrench to tighten evenly, alternate from one nut to the other until reaching the recommended torque.

2. When two clips are required, apply the second clip as near the loop or thimble as possible. Use torque wrench to tighten evenly, alternating until reaching the recommended torque. When more than two clips are required, apply the second clip as near the loop or thimble as possible, turn nuts on second clip firmly, but do not tighten. Proceed to Step 3.



Figure 2

3. When three or more clips are required, space additional clips equally between first two – take up rope slack – use torque wrench to tighten nuts on each U-Bolt evenly, alternating from one nut to the other until reaching recommended torque.



Figure 3

4. If a pulley (sheave) is used in place of a thimble, add one additional clip. Clip spacing should be as shown.

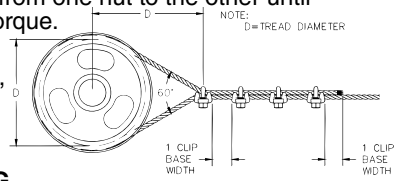


Figure 4

5. **WIRE ROPE SPLICING PROCEDURES:**

The preferred method of splicing two wire ropes together is to use inter-locking turnback eyes with thimbles, using the recommended number of clips on each eye (See Figure 5).



Figure 5

An alternate method is to use twice the number of clips as used for a turnback termination. The rope ends are placed parallel to each other, overlapping by twice the turnback amount shown in the application instructions. The minimum number of clips should be installed on each dead end (See Figure 6). Spacing, installation torque, and other instructions still apply.

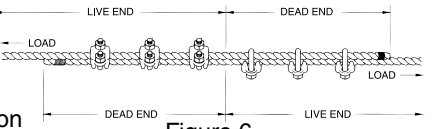


Figure 6

6. **IMPORTANT**

Apply first load to test the assembly. This load should be of equal or greater weight than loads expected in use. Next, check and use torque wrench to retighten to recommended torque. In accordance with good rigging and maintenance practices, the wire rope end termination should be inspected periodically for wear, abuse, and general adequacy.

Table 1

Clip Size (in.)	Rope Size (in.)	Minimum No. of Clips	Amount of Rope to Turn Back in Inches	*Torque in Ft.Lbs.
1/8	1/8	2	3-1/4	4.5
3/16	3/16	2	3-3/4	7.5
1/4	1/4	2	4-3/4	15
5/16	5/16	2	5-1/4	30
3/8	3/8	2	6-1/2	45
7/16	7/16	2	7	65
1/2	1/2	3	11-1/2	65
9/16	9/16	3	12	95
5/8	5/8	3	12	95
3/4	3/4	4	18	130
7/8	7/8	4	19	225
1	1	5	26	225
1-1/8	1-1/8	6	34	225
1-1/4	1-1/4	7	44	360
1-3/8	1-3/8	7	44	360
1-1/2	1-1/2	8	54	360
1-5/8	1-5/8	8	58	430
1-3/4	1-3/4	8	61	590
2	2	8	71	750
2-1/4	2-1/4	8	73	750
2-1/2	2-1/2	9	84	750
2-3/4	2-3/4	10	100	750
3	3	10	106	1200
3-1/2	3-1/2	12	149	1200
If a pulley (sheave) is used for turning back the wire rope, add one additional clip. See Figure 4.				
If a greater number of clips are used than shown in the table, the amount of turnback should be increased proportionately.				
*The tightening torque values shown are based upon the threads being clean, dry, and free of lubrication.				

Copyright © 2014 The Crosby Group LLC
All Rights Reserved

Fist Grip® Wire Rope Clips

SEE APPLICATION AND WARNING INFORMATION
Para Español: www.thecrosbygroup.com *on Page 121*

NEW STYLE FIST GRIP® CLIPS 3/16″ - 5/8″



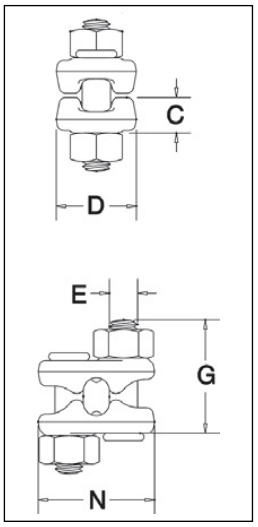
Fist Grip® wire clips meet or exceed the performance requirements of Federal Specification FF-C-450 Type III, Class 1, except for those provisions required of the contractor. For additional information, see Crosby General Catalog.

- Entire clip is galvanized to resist corrosive and rusting action.
- Based on the catalog breaking strength of wire rope, Crosby wire rope clips have an efficiency rating of 80% for 3/16″ - 7/8″ sizes, and 90% for sizes 1″ through 1-1/2″.
- Bolts are an integral part of the saddle. Nuts can be installed in such a way as to enable the operator to swing the wrench in a full arc for fast installation.
- All sizes have forged steel saddles.
- All clips are individually bagged or tagged with proper application instructions and warning information.
- Meets or exceeds all requirements of ASME B30.26 including identification, ductility, design factor, proof load and temperature requirements. Importantly, these wire rope clips meet other critical performance requirements including fatigue life, impact properties and material traceability, not addressed by ASME B30.26.
- Assembled with standard heavy hex nuts.

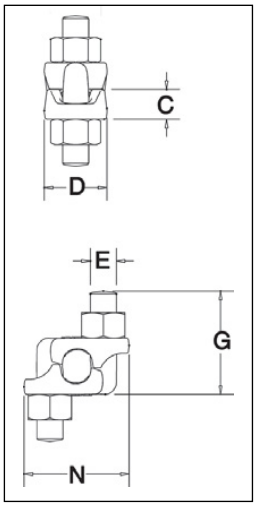
FIST GRIP® CLIPS 3/4″ - 1-1/2″



3/16″ - 5/8″



3/4″ - 1-1/2″



G-429 Fist Grip® Clips

Rope Size (in.)*	Rope Size (mm)	G-429 Stock No.	Std. Package Qty.	Weight Per 100 (lbs.)	Dimensions (in.) C	D	E	G	N
3/16 - 1/4	5-7	1010471	100	23	.40	.94	.38	1.41	1.44
5/16	8	1010499	100	28	.47	1.06	.38	1.50	1.54
3/8	10	1010514	50	40	.51	1.06	.44	1.84	1.78
7/16 - 1/2	11-13	1010532	50	62	.59	1.25	.50	2.21	2.15
9/16 - 5/8	14-16	1010550	50	103	.72	1.50	.63	2.72	2.57
3/4	18-20	1010578	25	175	.86	1.81	.75	2.94	2.67
7/8	22	1010596	25	225	.97	2.12	.75	3.31	2.86
1	24-26	1010612	10	300	1.13	2.25	.75	3.72	3.06
1-1/8	28-30	1010630	10	400	1.28	2.38	.88	4.22	3.44
1-1/4	32-34	1010658	10	400	1.34	2.50	.88	4.25	3.56
1-3/8 - 1-1/2	36-40	1010676	Bulk	700	1.56	3.00	1.00	5.56	4.12

* Sizes through 5/8″ incorporate New Style Design.

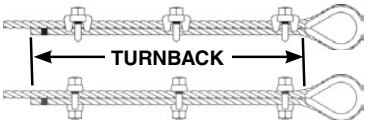
Copyright © 2014 The Crosby Group LLC
All Rights Reserved

Wire Rope Clip Information

Fist Grip® Clip Installation Procedures

WIRE ROPE CLIPS

BASIC REQUIREMENTS FOR PROPER TERMINATION



SELECT PROPER SIZE CLIPS AND TURNBACK.

PLACE CLIPS ON IN PROPER SEQUENCE.

TORQUE ALL CLIPS EVENLY WITH A TORQUE WRENCH.

APPLY FIRST LOAD AND RE-TORQUE WITH A TORQUE WRENCH.

(POH651)

Turnback is measured from the end of the rope to the base of the eye or to the thimble. This part of the rope is often referred to as the dead end or "dead horse". Since the Fist Grip® clip utilizes two saddles, you do not have to be concerned with workers saddling the dead horse.

The table below defines the information required for each wire rope clip to achieve maximum efficiency.

Table 1

Clip Size (in.)	Rope Size (in.)	Minimum No. of Clips	Amount of Rope to Turn Back in Inches	*Torque in Ft. Lbs.
3/16	3/16	2	4	30
1/4	1/4	2	4	30
5/16	5/16	2	5	30
3/8	3/8	2	5-1/4	45
7/16	7/16	2	6-1/2	65
1/2	1/2	3	11	65
9/16	9/16	3	12-3/4	130
5/8	5/8	3	13-1/2	130
3/4	3/4	3	16	225
7/8	7/8	4	26	225
1	1	5	37	225
1-1/8	1-1/8	5	41	360
1-1/4	1-1/4	6	55	360
1-3/8	1-3/8	6	62	500
1-1/2	1-1/2	7	78	500
If a pulley (sheave) is used for turning back the wire rope, add one additional clip. See Figure 4.				
If a greater number of clips are used than shown in the table, the amount of turnback should be increased proportionately.				
* The tightening torque values shown are based upon the threads being clean, dry and free of lubrication.				

The following example illustrates the correct use of Table 1 in determining the proper application of Crosby G-429 Fist Grip® Clips.

Example: An installation requires the use of 7/16" 6 x 19 RRL, IPS wire rope.

From Table 1:

– Clip Size = 7/16"
– Number of Clips = 2
– Amount of Turnback = 6-1/2"
– Torque in Ft. Lbs. = 65

Table 1

Clip Size (in.)	Rope Size (in.)	Minimum No. of Clips	Amount of Rope to Turn Back in Inches	*Torque in Ft. Lbs.
1/8	1/8	2	3-1/4	4.5
3/16	3/16	2	3-3/4	7.5
1/4	1/4	2	4-3/4	15
5/16	5/16	2	5-1/4	30
3/8	3/8	2	6-1/2	45
7/16	7/16	2	7	65

Copyright © 2014 The Crosby Group LLC
All Rights Reserved

CROSBY® FIST GRIP® CLIPS

WARNINGS AND APPLICATION INSTRUCTIONS



New Style Fist Grip®
3/16" - 5/8"



Fist Grip® Clips
3/4" - 1-1/2"

⚠ WARNING

- **Failure to read, understand, and follow these instructions may cause death or serious injury.**
- **Read and understand these instructions before using clips.**
- **Match the same size clip to the same size wire rope.**
- **Do not mismatch Crosby clips with other manufacturer's clips.**
- **Prepare wire rope end termination only as instructed.**
- **Do not use with plastic coated wire rope.**
- **Apply first load to test the assembly. This load should be of equal or greater weight than loads expected in use. Next, check and retighten nuts to recommended torque (See Table 1, this page).**

Efficiency ratings for wire rope end terminations are based upon the catalog breaking strength of wire rope. The efficiency rating of a properly prepared loop or thimble-eye termination for clip sizes 1/8" through 7/8" is 80%, and for sizes 1" through 3-1/2" is 90%.

The number of clips shown (see Table 1) is based upon using RRL or RLL wire rope, 6 x 19 or 6 x 37 Class, FC or IWRC; IPS or XIP, XXIP. If Seale construction or similar large outer wire type construction in the 6 x 19 Class is to be used for sizes 1 inch and larger, add one additional clip. If a pulley (sheave) is used for turning back the wire rope, add one additional clip.

The number of clips shown also applies to rotation-resistant RRL wire rope, 8 x 19 Class, IPS, XIP, XXIP sizes 1-1/2 inch and smaller; and to rotation-resistant RRL wire rope, 19 x 7 Class, IPS, XIP, XXIP sizes 1-3/4 inch and smaller.

For other classes of wire rope not mentioned above, we recommend contacting Crosby Engineering at the address or telephone number on the back cover to ensure the desired efficiency rating.

The style of wire rope termination used for any application is the obligation of the user.

The style wire rope termination used for any application is the obligation of the user.

For OSHA (Construction) applications, see OSHA 1926.251.

1. Refer to Table 1 in following these instructions. Turn back specified amount of rope from thimble or loop. Apply first clip one base width from dead end of rope. Use torque wrench to tighten evenly, alternating from one nut to the other until reaching the recommended torque.

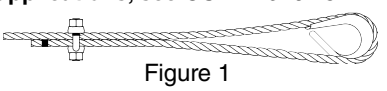


Figure 1

2. When two clips are required, apply the second clip as near the loop or thimble as possible. Use torque wrench to tighten evenly, alternating until reaching the recommended torque. When more than two clips are required, apply the second clip as near the loop or thimble as possible, turn nuts on second clip firmly, but do not tighten. Proceed to Step 3.



Figure 2

3. When three or more clips are required, space additional clips equally between first two – take up rope slack – use torque wrench to tighten on each Clip evenly, alternating from one nut to the other until reaching recommended torque.

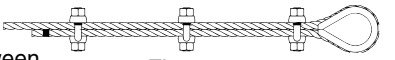


Figure 3

4. If a pulley (sheave) is used in place of a thimble, add one additional Fist Grip. Fist Grip spacing should be as shown.

5. **WIRE ROPE SPLICING PROCEDURES:**

The preferred method of splicing two wire ropes together is to use inter-locking turnback eyes with thimbles, using the recommended number of clips on each eye (See Figure 5).

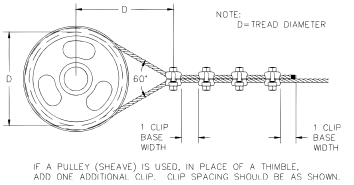


Figure 4

An alternate method is to use twice the number of clips as used for a turnback termination. The rope ends are placed parallel to each other, overlapping by twice the turnback amount shown in the application instructions. The minimum number of clips should be installed on each dead end (See Figure 6). Spacing, installation torque, and other instructions still apply.



Figure 5

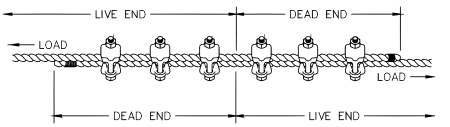


Figure 6

6. **IMPORTANT**

Apply first load to test the assembly. This load should be of equal or greater weight than loads expected in use. Next, check and use torque wrench to retighten to recommended torque.

In accordance with good rigging and maintenance practices, the wire rope end termination should be inspected periodically for wear, abuse, and general adequacy.

Table 1

Clip Size (in.)	Rope Size (in.)	Minimum No. of Clips	Amount of Rope to Turn Back in Inches	*Torque in Ft.Lbs.
3/16	3/16	2	4	30
1/4	1/4	2	4	30
5/16	5/16	2	5	30
3/8	3/8	2	5-1/4	45
7/16	7/16	2	6-1/2	65
1/2	1/2	3	11	65
9/16	9/16	3	12-3/4	130
5/8	5/8	3	13-1/2	130
3/4	3/4	3	16	225
7/8	7/8	4	26	225
1	1	5	37	225
1-1/8	1-1/8	5	41	360
1-1/4	1-1/4	6	55	360
1-3/8	1-3/8	6	62	500
1-1/2	1-1/2	7	78	500
If a pulley (sheave) is used for turning back the wire rope, add one additional clip. See Figure 4.				
If a greater number of clips are used than shown in the table, the amount of turnback should be increased proportionately.				
*The tightening torque values shown are based upon the threads being clean, dry, and free of lubrication.				

Copyright © 2014 The Crosby Group LLC
All Rights Reserved

Notes

Copyright © 2014 The Crosby Group LLC
All Rights Reserved

Notes

Copyright © 2014 The Crosby Group LLC
All Rights Reserved